WILDFIRES
IN CALIFORNIA

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https://docs.google.com/presentation/d/14PZSPLSzN30shtIZBn17ukswLaJj0469xb0ddrG1vsk/edit?usp=sharing

DATASCI 207 Final Presentation

THE QUESTION

How can ML models improve short term wildfire prediction in California?

Why this matters

Supports Early Warning

Planning for Resource Allocation

Scalability & Expansion

Improves Policy Decisions

EXISTING WILDFIRE ML MODELS

Item	Paper	Model	Features	Prediction
Data-Driven Wildfire Risk Prediction in Northern California (atmosphere 2021)	Atmosphere 2021	Random Forest - 92%, Adabost - 91.5%, Gradient Boosting Trees - 90.5%	Weather, Terrain, Powerline and Vegetation	Fire / No Fire
Identifying Key Drivers of Wildfires in the contiguous US using Machine Learning and Gaming Theory (Earth's Future, May 2021)	Earth's Future, May 2021	eXtreme Gradient Boosting Model - RMSE 2.04 km squared	Local Meteorology, Large Scale Meteorological Patterns, Land Surface Properties and Socio-Economic	Size of Burnt Area
Wildfire Prediction Through Live Fuel Moisture Content Maps (Civil and Environmental Engineering, Stanford University)	Civil and Environmental Engineering, Stanford University	SVM - 65.86%, Random Forest - 71.95%, CNN - 52.46%	Live Fuel Moisture Content (LFMC) Maps	Fire / No Fire

Reference: Ritter, J., Young, M., & Krishnan, P. (2022, August 4). *California wildfires prediction* [Final presentation]. W207 Final Project. [https://corneliailin.github.io/datasci_w207_spring2025/summer2022_projects/Prakash_Krishnan_Final_Presentation_CA_Wildfires_Aug_4_2022.pdf]

APPROACH

1. Develop Research Question
2. Data Selection
3. EDA Pre-processing
4. Model Selection
5. Build Baseline
6. Location Data Selection
7. Data Merges
8. EDA
9. Pre-processing (inc. Temporal Split)
10. Feature Engineering
11. Experimentation & Measurement
12. Conclusion

THE DATA

Original Data
California Weather & Fire Prediction:
https://zenodo.org/records/14712845

Size: 14,988 rows, 14 columns
Timeframe: 01-01-1984 to 01-12-2025

Geolocated Data
Cal Historical Fire: All & 2024/2025
https://www.fire.ca.gov/our-impact/statistics

NASA POWER : Localized averaged weather data
https://power.larc.nasa.gov/data-access-viewer/

USA Counties Shapefile:
https://www2.census.gov/geo/tiger/TIGER2024/COUNTY/tl_2024_us_county.zip

USA CONUS Climate Divisions Shapefile
https://www.ncei.noaa.gov/access/monitoring/reference-maps/conus-climate-divisions

Final Size of Merged Data: 20,710 rows, 108 columns
Timeframe: 01-01-1984 to 01-12-2025

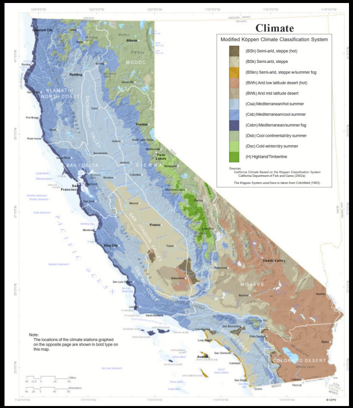
INDEX:
DATE

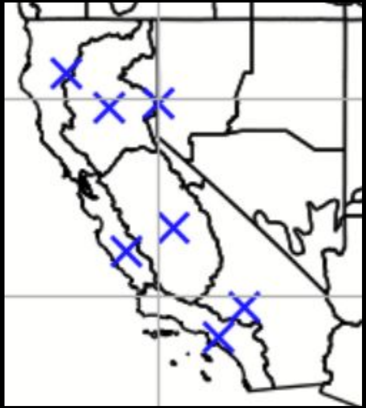
LABELS:
CONUS CODE

NUMERICAL:
Precipitation
Min/Max Temp
(Daily) Temp Range
Avg Wind Speed
Wind Temp. Ratio
Lagged Precip.
Lagged Avg Wind Speed
Lagged Temp

TARGET VARIABLES:
Fire in 3 Days
Fire in 7 Days
Fire in 14 Days

THE DATA



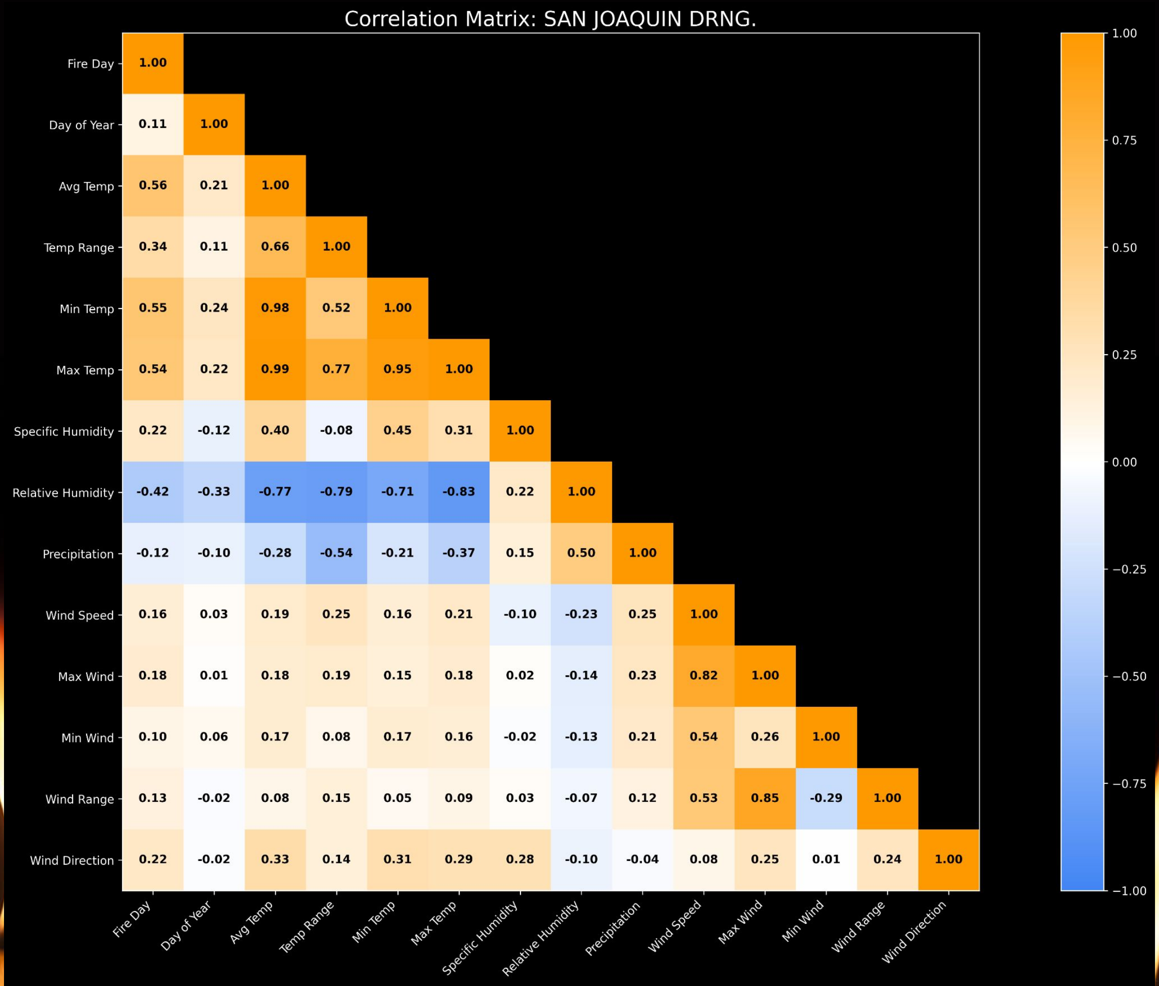


1	North Coast Drainage
2	Sacramento Drainage
3	Northeast Interior Basins
4	Central Coast Drainage
5	San Joaquin Drainage
6	South Coast Drainage
7	Southeast Desert Basin

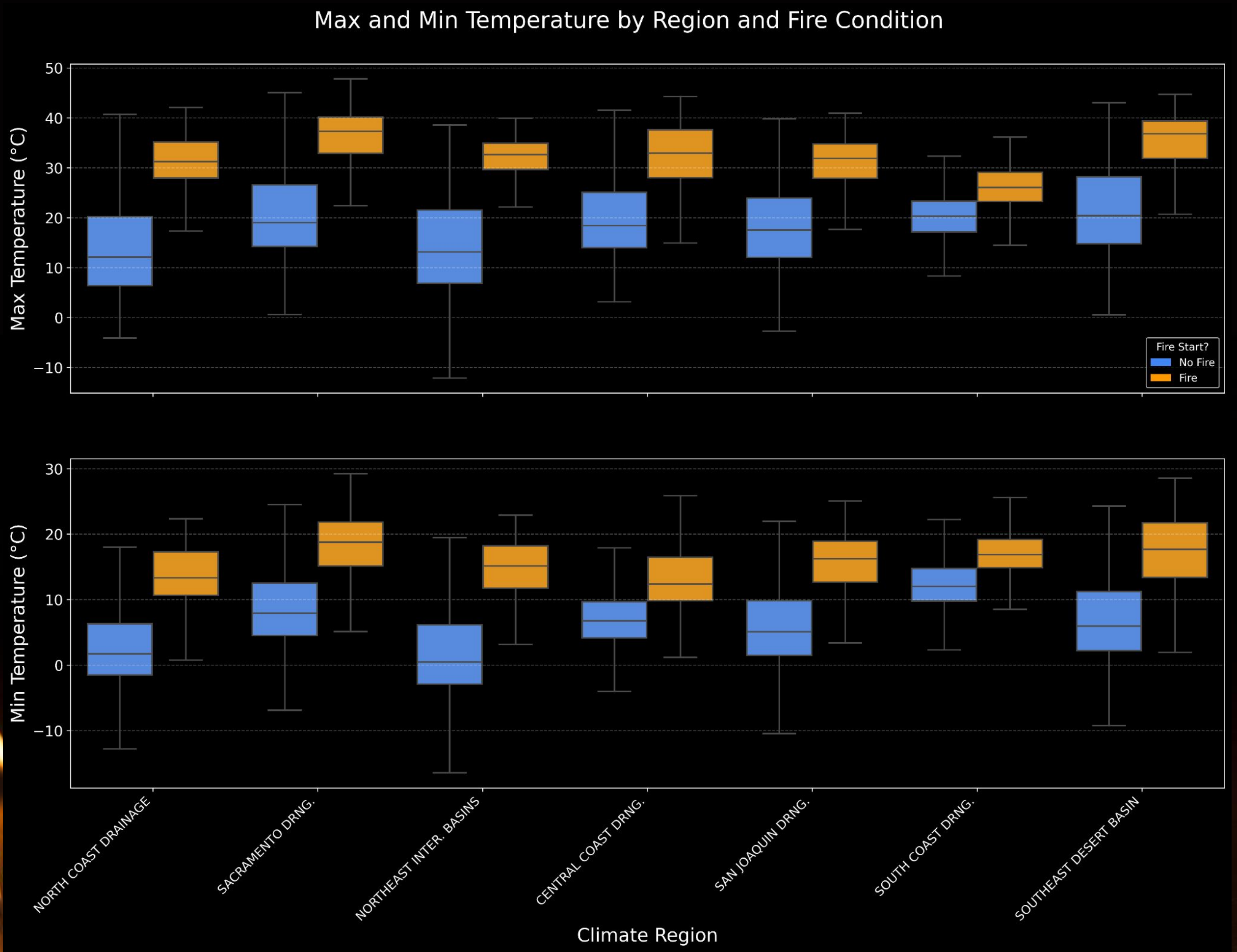
*Shapely" to find one fire centroid per climate division.

*GEOPandas" to geo-locate labels and features to counties and climate divisions

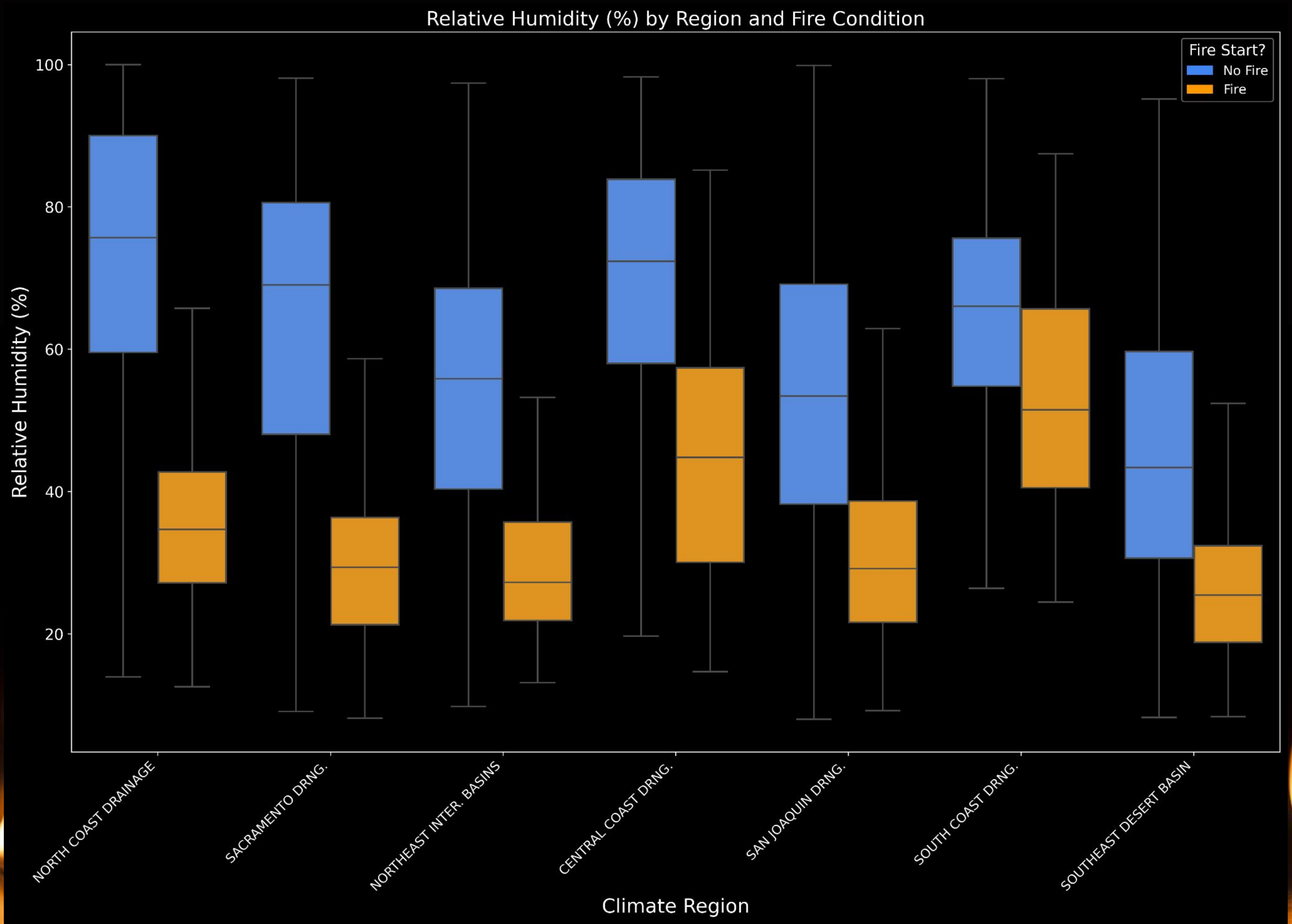
EDA - FEATURE CORRELATION



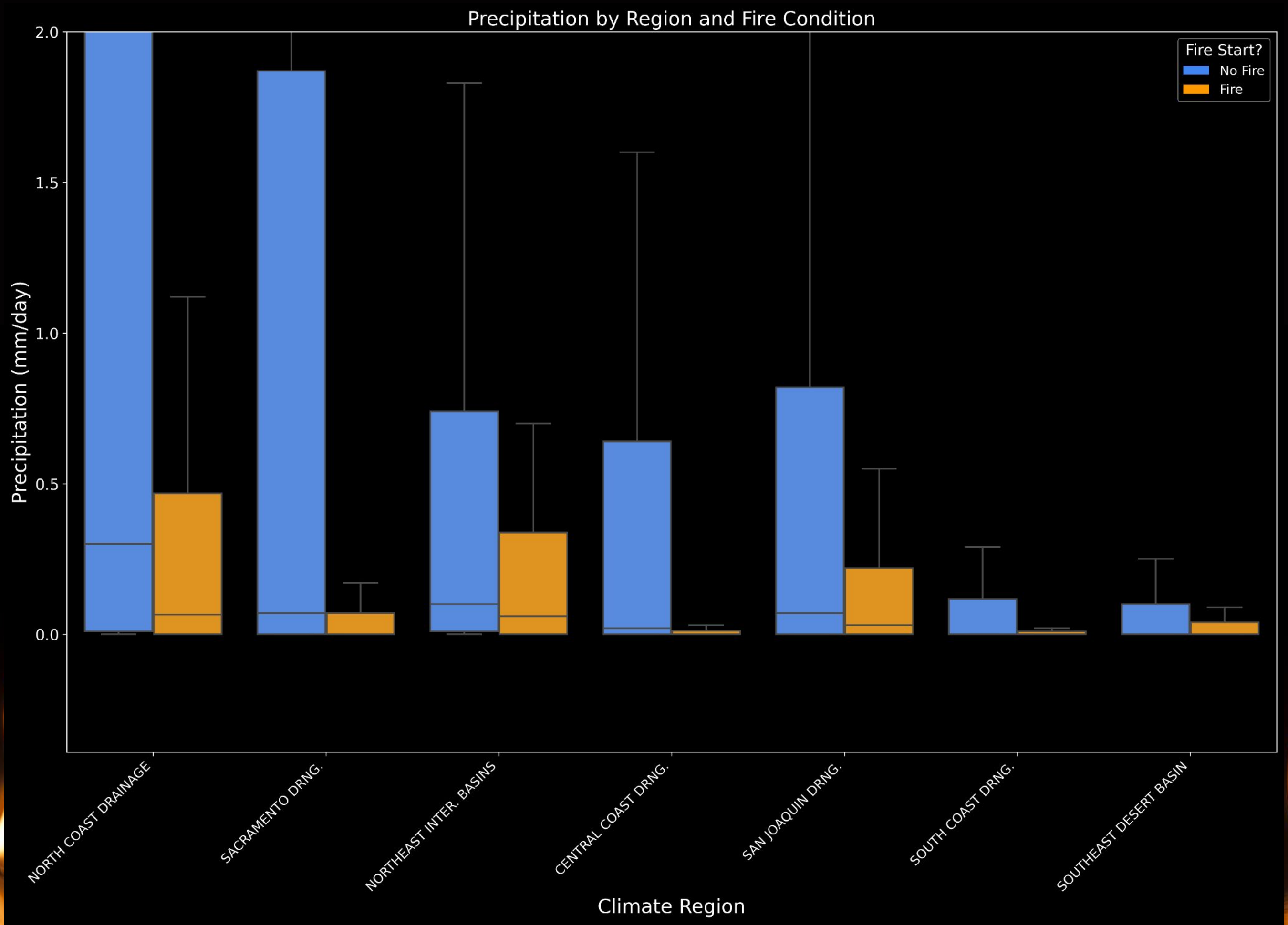
EDA - TEMPERATURE



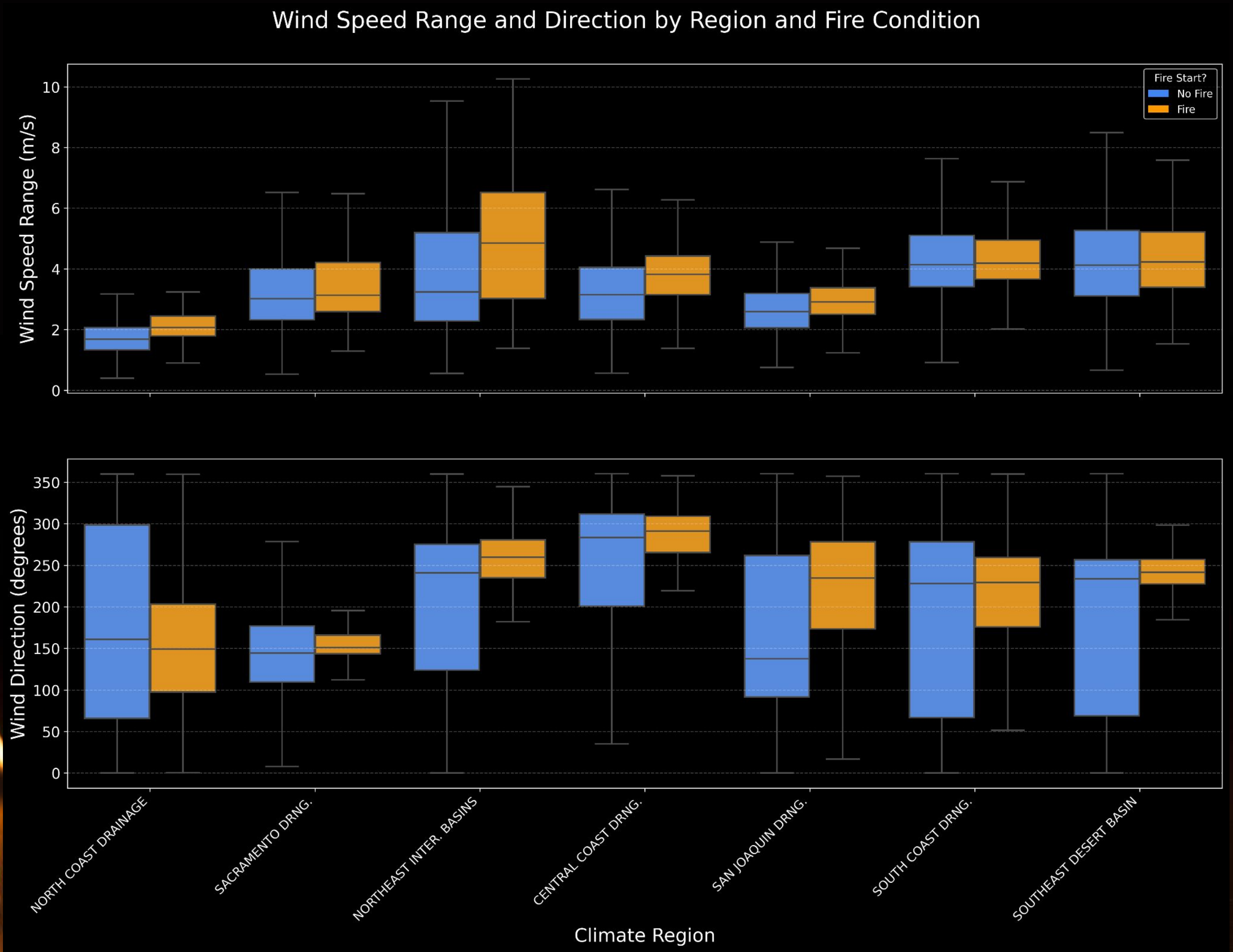
EDA - RELATIVE HUMIDITY



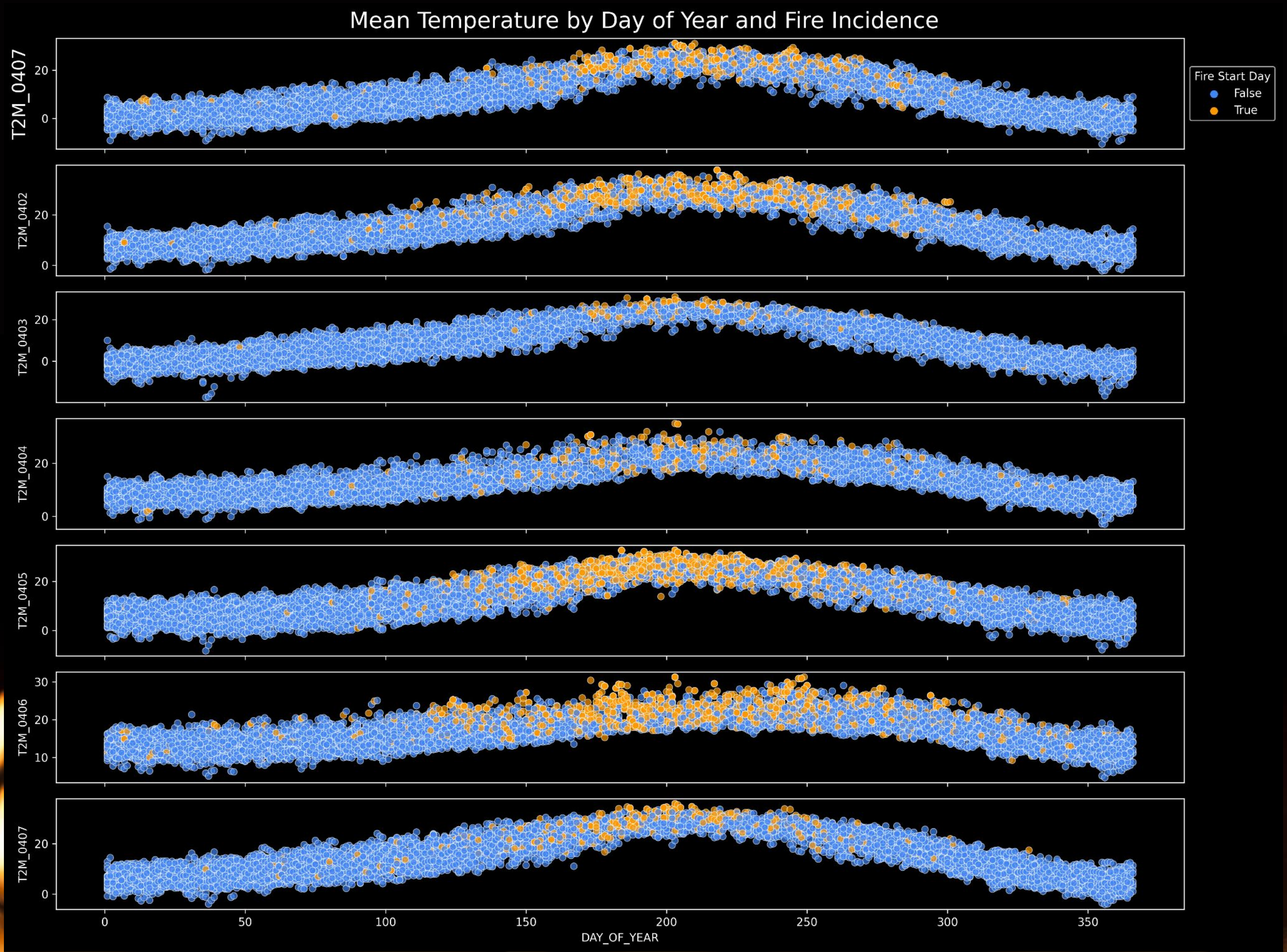
EDA - PRECIPITATION



EDA - WIND SPEED AND DIRECTION



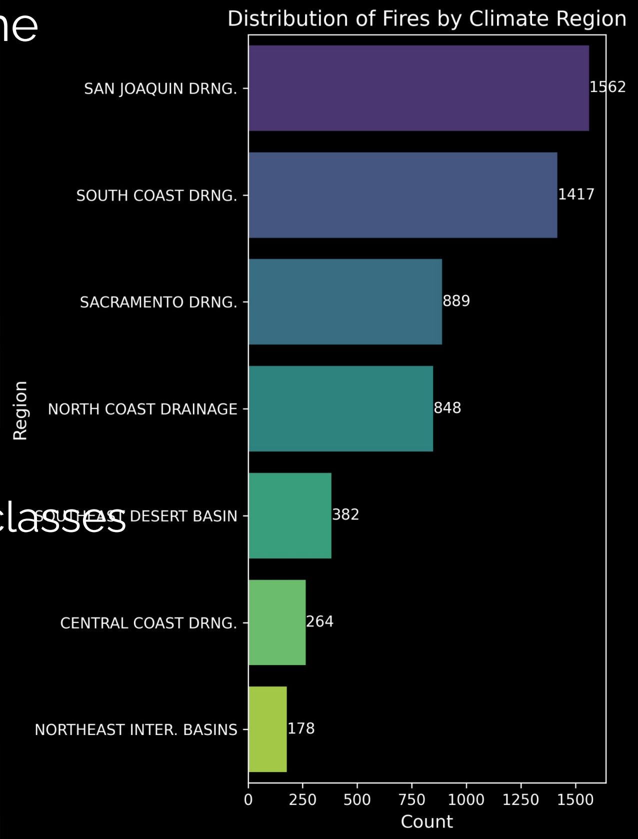
EDA - SEASONALITY



EDA - CLASS IMBALANCE

- Slight imbalance in training dataset binary outcome (64% days without a fire), moderate risk of:
  - High accuracy but lower recall
  - overfitting

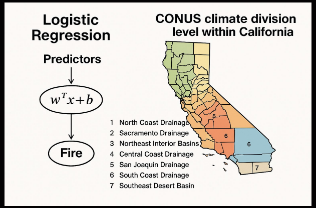
- Fire prevalence varies widely by climate region
  - 178 examples in Northeast Inter. Basins
  - 1,562 examples in San Joaquin Drainage
  - Nearly 9-fold difference b/w smallest and largest classes



EXPERIMENT SUMMARY: BASELINE AND ADVANCED

Item #	Purpose	ML Model	Test Evaluation Metric	% Improve Over Baseline	Features and Labels	Hyper Parameters
1	Short-term wildfire outbreak prediction model for California (3, 7, and 14-day forecasts) Targets outbreak prediction at the CONUS climate division level within California	Baseline Logistic regression (SciKit Learn)	Accuracy Precision Recall F1 y_3 0.453 0.506 0.453 0.473 y_7 0.320 0.404 0.320 0.349 y_14 0.232 0.318 0.232 0.260	Not Applicable	**Labels:** 3, 7, 14 days forecasts, CONUS climate divisions **Features:** Geolocated Fire and weather data	Algo: Multinomial Logistic Regression Regularization Strength: C=1.0 Solver: LBFGS Class Weighting: Balanced Max Iterations: 1000 Multi-class Strategy: Multinomial None tuned
2	Short-term wildfire outbreak prediction model for California (3, 7, and 14-day forecasts) Targets outbreak prediction at the CONUS climate division level within California	Feedforward Neural Network with Tuned hyperparameter selections (Optuna)	Accuracy Precision Recall F1 y_3 0.508 0.531 0.508 0.517 y_7 0.367 0.417 0.367 0.381 y_14 0.257 0.328 0.257 0.275	3-day: 12.1% 7-day: 14.7% 14-day: 10.8%	**Labels:** 3, 7, 14 days forecasts, CONUS climate divisions **Features:** Geolocated Fire and weather data	n_layers, n_units, activation, dropout, learning_rate, batch_size, lr_decay_factor, lr_patience Tuned using Optuna
3	Short-term wildfire outbreak prediction model for California (3, 7, and 14-day forecasts) Targets outbreak prediction at the CONUS climate division level within California	Random Forest Random Forest + PCA RF + SMOTE + PCA	Accuracy Precision Recall F1 y_3 0.533 0.470 0.422 0.438 Y_7 0.405 0.327 0.288 0.290 y_14 0.341 0.232 0.229 0.217	3-day: 8% 7-day: 8.5% 14-day: 10%	**Labels:** 3, 7, 14 days forecasts, CONUS climate divisions **Features:** Geolocated Fire and weather data	n_estimators,max_depth, min_samples_split, min_samples,max_features , class_weight, random_state, n_jobs
4	Short-term wildfire outbreak prediction model for California (3, 7, and 14-day forecasts) Targets outbreak prediction at the CONUS climate division level within California	XGBoost with random search for hyperparameter tuning	Accuracy Precision Recall F1 y_3 0.542 0.525 0.542 0.525 y_7 0.406 0.393 0.406 0.369 y_14 0.338 0.347 0.338 0.284	3-day: 8.9% 7-day: 8.6% 14-day: 10.6%	**Labels:** 3, 7, 14 days forecasts, CONUS climate divisions **Features:** Geolocated Fire and weather data	n_estimators, max_depth, learning_rate, reg_alpha, reg_lambda, min_child_weight, gamma, subample, colsample_bytree
5	Short-term wildfire outbreak prediction model for California (3, 7, and 14-day forecasts) Targets outbreak prediction at the CONUS climate division level within California	ARIMA (SARIMAX)	Accuracy Precision Recall F1 y_3 0.672 0.558 0.690 0.617 y_7 0.659 0.529 0.692 0.600 y_14 0.650 0508 0.695 0.587	3-day: 21.9% 7-day:33.9% 14-day: 41.8%	**Labels:** 3, 7, 14 days forecasts, CONUS climate divisions **Features:** Geolocated Fire and weather data	Order (p,d,q) Seasonal order (P, D, Q)

LOGISTIC REGRESSION (BASELINE) SCIKIT LEARN



Key Insights

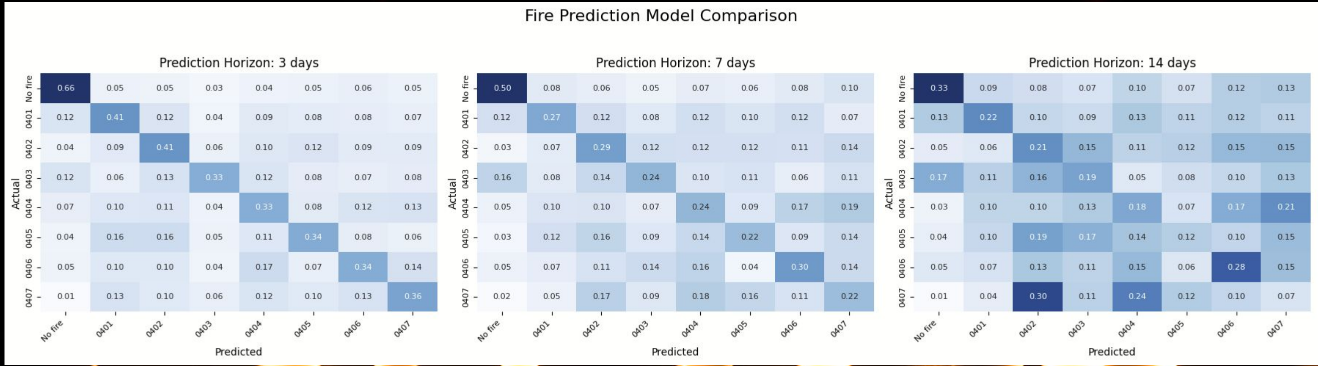
- Performance degrades with longer horizons, yet even 14-day predictions (23.2% accuracy) outperform random guessing by 85% in our 8-class problem, confirming genuine predictive power.
- Higher precision than accuracy indicates the model selectively predicts fires, reducing false alarms while maintaining detection capability.
- The model delivers strong short-term forecasting (3-day: 45.3% accuracy) while still providing actionable intelligence for longer horizons despite increasing uncertainty.

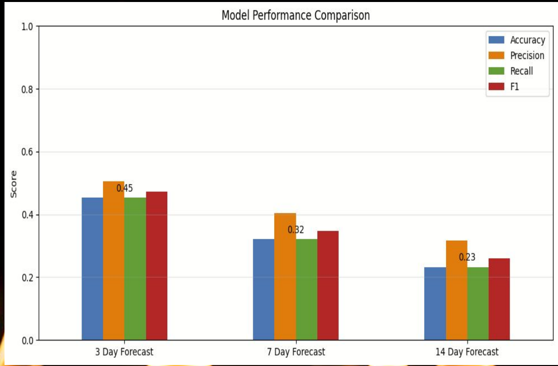
Hyperparameters

- Algorithm: Multinomial Logistic Regression
- Regularization Strength: C=1.0 (moderate)
- Solver: LBFGS (quasi-Newton method)
- Class Weighting: Balanced (inversely proportional to class frequencies)
- Max Iterations: 1000
- Multi-class Strategy: Multinomial (true softmax-based approach)

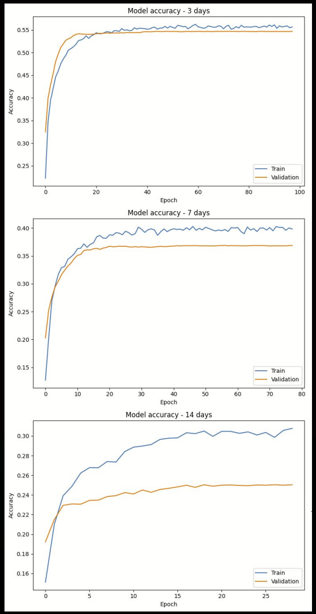
Performance Metrics

	Accuracy	Precision	Recall	F1
y_3	0.453	0.506	0.453	0.473
y_7	0.320	0.404	0.320	0.349
y_14	0.232	0.318	0.232	0.260





NEURAL NETWORK

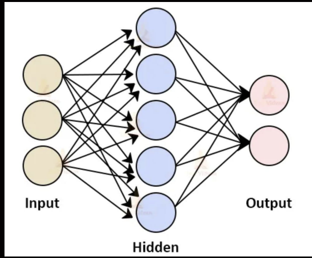


FF Neural Network vs. Baseline: Key Findings

- FF Neural networks outperform logistic regression across all prediction horizons, with the most significant improvement in 3-day forecasts (+5.5 points, reaching 50.8% accuracy).
- Performance degrades with longer horizons in both models, but neural networks degrade more gracefully, providing 10.8% better accuracy for 14-day predictions.
- Improved precision (3-5%) indicates fewer false alarms, while enhanced recall means more actual fire events are correctly identified - critical for emergency response planning.
- The optimized neural architecture successfully captures complex meteorological patterns that the simpler baseline model cannot detect.

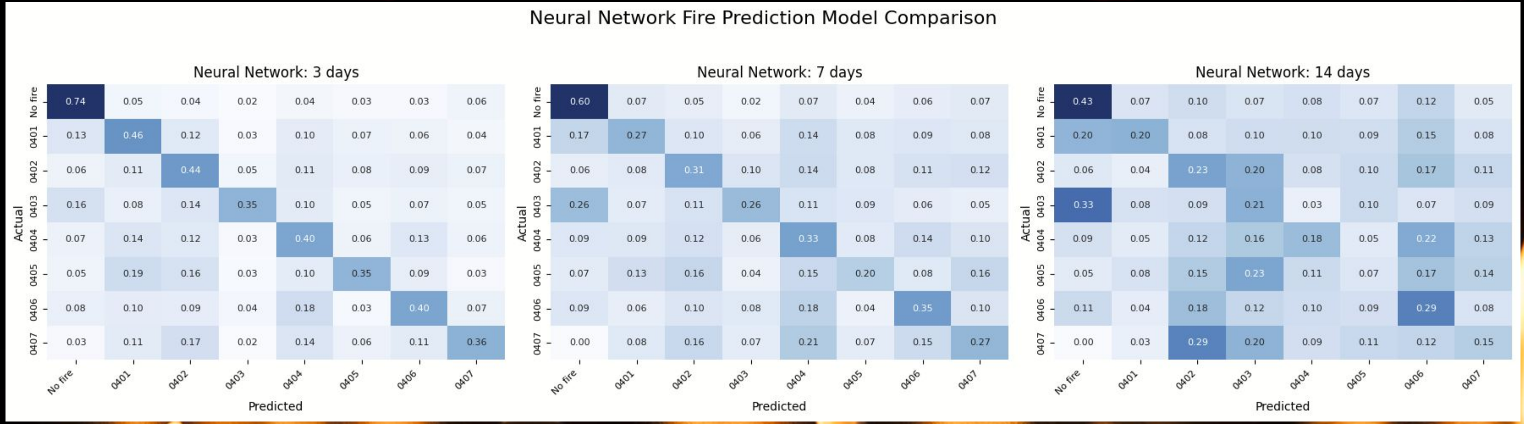
Best hyperparameters:

- batch_size: 128
- n_layers: 3
- n_units_l0: 508
- activation_l0: selu
- dropout_l0: 0.497968874445551813
- n_units_l1: 45
- activation_l1: selu
- dropout_l1: 0.003571962578178628
- n_units_l2: 367
- activation_l2: elu
- dropout_l2: 0.1540127137562558
- learning_rate: 1.3230185409824418e-05
- lr_decay_factor: 0.49956238105043055
- lr_patience: 3
  - Best validation accuracy: 0.5505



Experiments:

- Optuna for all hyperparameters
- Weights for classes
- Lagged features
- Other architectures: CNN, LSTM



	Accuracy	Precision
y_3	0.508	0.531
y_7	0.367	0.417
y_14	0.257	0.328

	Recall	F1
y_3	0.508	0.517
y_7	0.367	0.381
y_14	0.257	0.275

Key Insights

- 10% increased accuracy over baseline logistic model for predicting that a fire starts within 3, 7, and 14 days.
- Feature importance insights highlight temperature and humidity as key predictors
- Handles non-linearity between fire occurrence and weather variables

RANDOM FOREST

Hyperparameters

- Algorithm: RandomForestClassifier
- n_estimators: 100
- max_depth: 20
- random_state: 42
- class_weight: 'balanced'
- n_jobs: -1

Best tuned hyperparameters

Model Summary

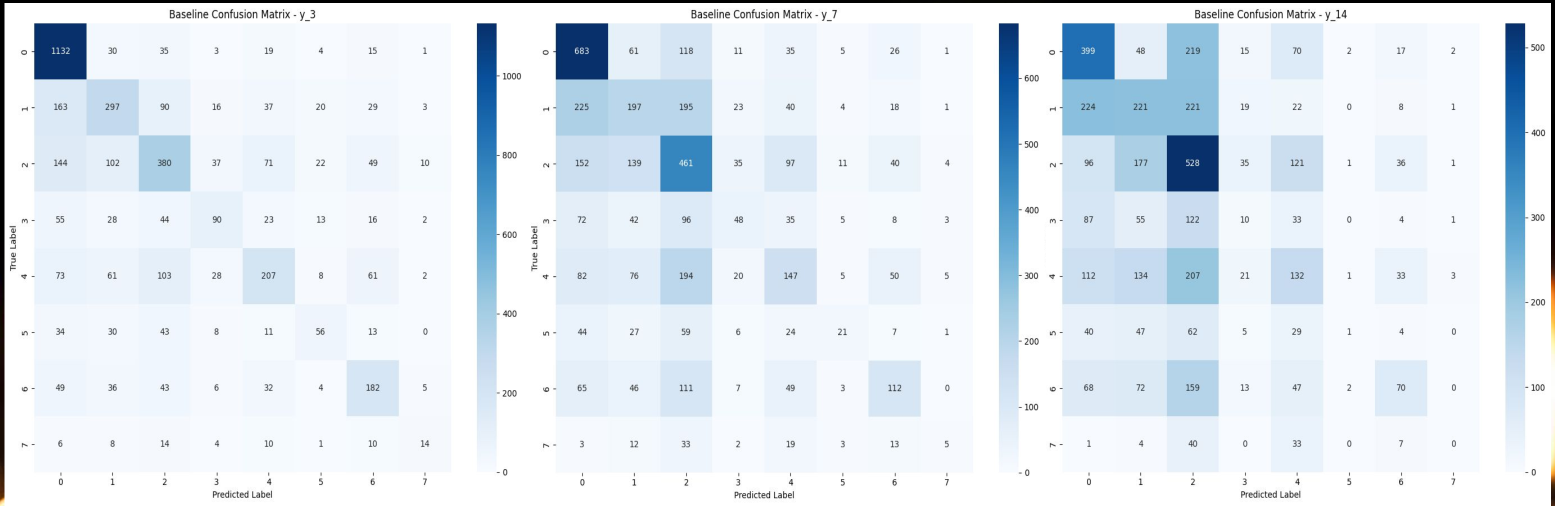
- Train Data: 12426 w/ 100+ features
- Validation Data: 4142 w/ 100+ features
- Test Data: 4142 w/ 100+ features

Best Parameters

- SMOTE sampling strategy: 'auto' (all classes balanced)
- n_estimators: 100
- max_depth: 20

Performance Metrics

	Accuracy	Precision	Recall	F1
y_3	0.533	0.470	0.422	0.438
y_7	0.405	0.327	0.288	0.290
y_14	0.341	0.232	0.229	0.217



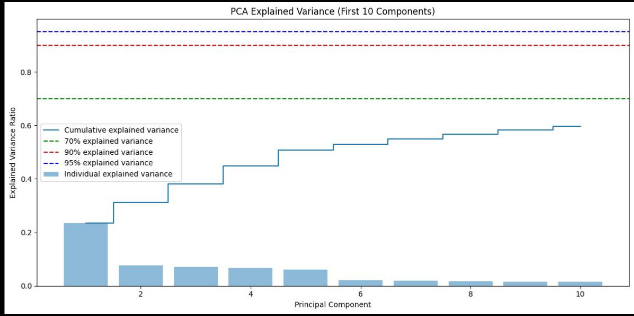
RANDOM FOREST + PCA

Key Insights

- Performed slightly worse than baseline RF
- Could be due to class imbalance and overfitting
- Applying sampling techniques such as SMOTE (Synthetic Minority Over-sampling Technique) could help

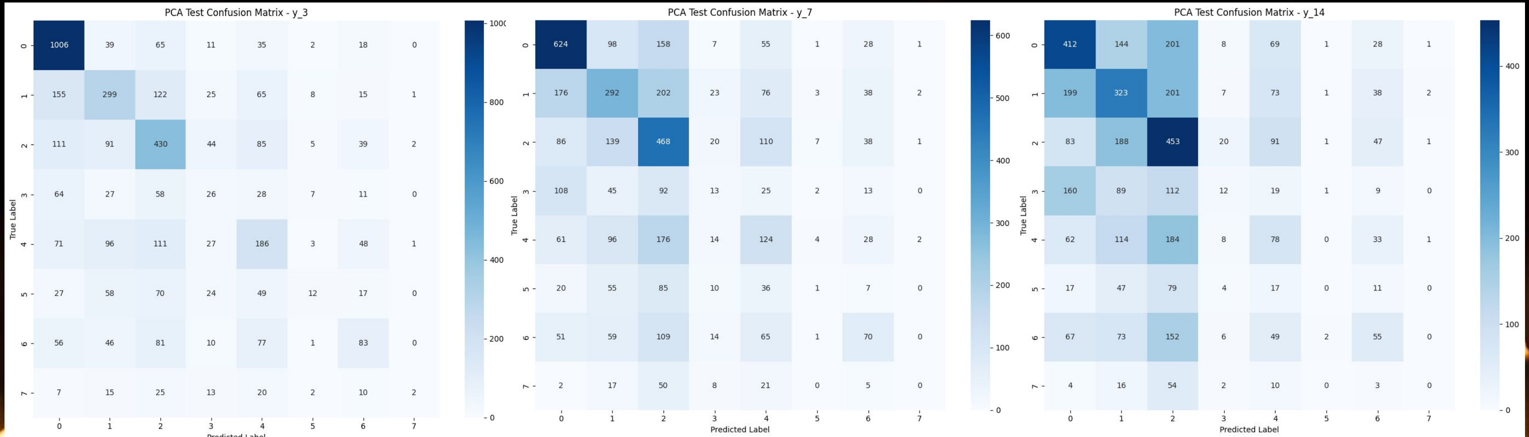
Hyperparameters

- Algorithm: RandomForestClassifier + PCA
- n_components: 10 for PCA
- n_estimators: 300
- max_depth: 50
- random_state: 42
- min_samples_split: 2
- min_samples_leaf: 1
- max_features: 'sqrt'
- class_weight: 'balanced'
- n_jobs: -1



Performance Metrics

	Accuracy	Precision	Recall	F1
y_3	0.494	0.367	0.319	0.313
y_7	0.387	0.246	0.251	0.238
y_14	0.324	0.211	0.215	0.200



RANDOM FOREST + SMOTE

Hyperparameters

- Algorithm: RandomForestClassifier + SMOTE
- SMOTE sampling strategy: 'auto' (all classes balanced)
- n_estimators: 100
- max_depth: 20
- Random state: 42 (for reproducibility)
- n_jobs: -1

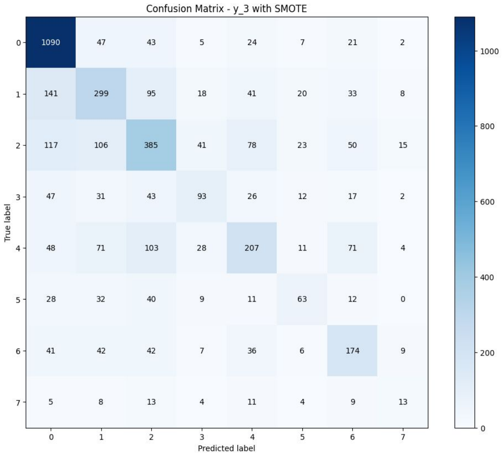
Key Insights

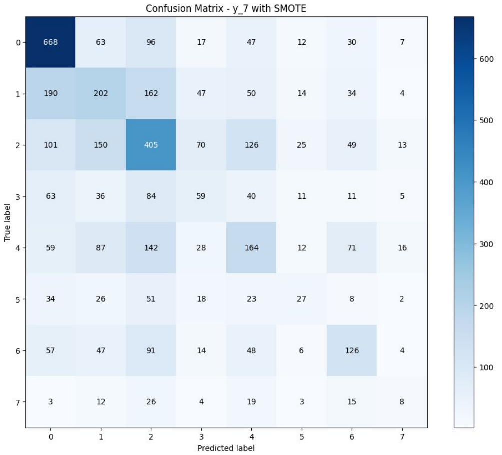
- Balanced dataset by creating synthetic examples for minority classes
- Improved recall for fire detection by over 23%
- Although accuracy is slightly less than baseline RF, improved overall in F1 score and Recall

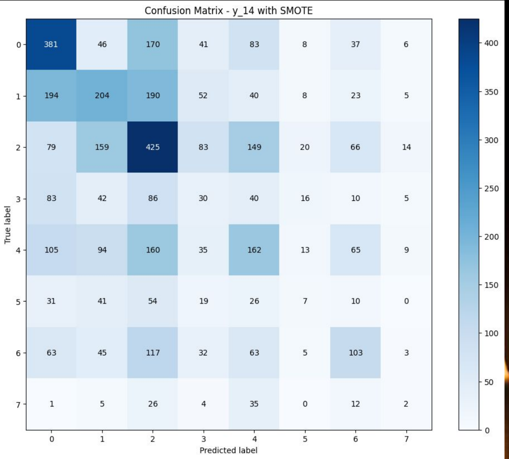
```
Class distribution before SMOTE for y_3:
    Class 0: 5196 samples (41.82%)
    Class 1: 1982 samples (15.95%)
    Class 2: 2133 samples (17.17%)
    Class 3: 566 samples (4.55%)
    Class 4: 1108 samples (8.92%)
    Class 5: 337 samples (2.71%)
    Class 6: 890 samples (7.16%)
    Class 7: 214 samples (1.72%)
Class distribution after SMOTE for y_3:
    Class 0: 5196 samples (12.50%)
    Class 1: 5196 samples (12.50%)
    Class 2: 5196 samples (12.50%)
    Class 3: 5196 samples (12.50%)
    Class 4: 5196 samples (12.50%)
    Class 5: 5196 samples (12.50%)
    Class 6: 5196 samples (12.50%)
    Class 7: 5196 samples (12.50%)
```

Performance Metrics

	Accuracy	Precision	Recall	F1
y_3	0.527	0.437	0.417	0.424
y_7	0.395	0.308	0.297	0.299
y_14	0.317	0.232	0.228	0.225







XGBOOST

Key Insights

- **All XGBoost models outperform baseline**, demonstrating strong pattern recognition at short and long horizons
- The **3-day model reaches 54.2% accuracy**, indicating high short-term predictive power
- The **7-day and 14-day models** show diminishing performance, reflecting the growing challenge of forecasting fire risk further into the future
- Precision remains relatively stable across time windows, showing the model remains **cautious and consistent in identifying true fire risks**

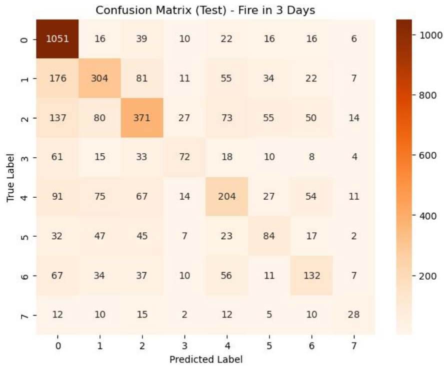
Performance Metrics

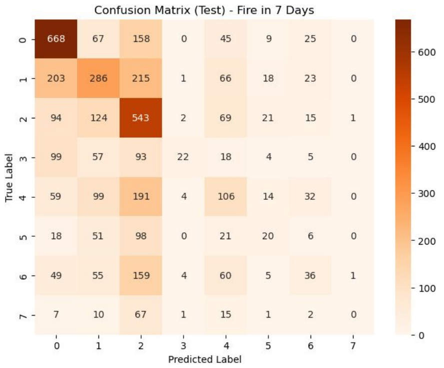
	Accuracy	Precision	Recall	F1
y_3	0.542	0.525	0.542	0.525
y_7	0.406	0.393	0.406	0.369
y_14	0.338	0.347	0.338	0.284

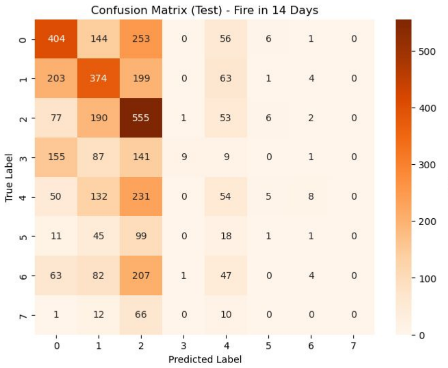
Hyperparameter Tuning

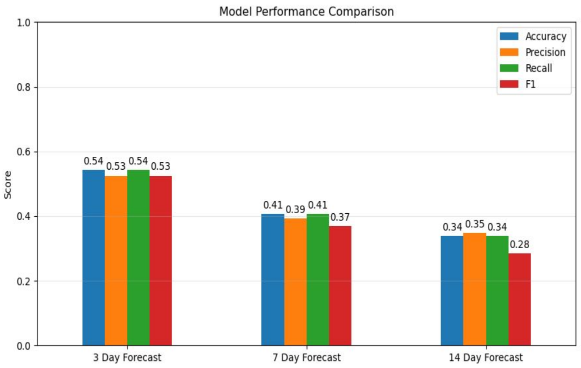
Random search with 20 iterations per model (best parameters listed below)

	n_estimators	max_depth	learning_rate	reg_alpha	reg_lambda	min_child_weight	gamma	subsample	colsample_bytree
3 Day	250	1	0.02	0.3	2.0	1.0	0.4	1.0	1.0
7 Day	200	5	0.04	0.3	1.0	5	0	0.8	0.3
14 Day	300	3	0.03	0.8	0.5	3	0.3	1.0	0.3









ARIMA/SARIMAX

Hyperparameters

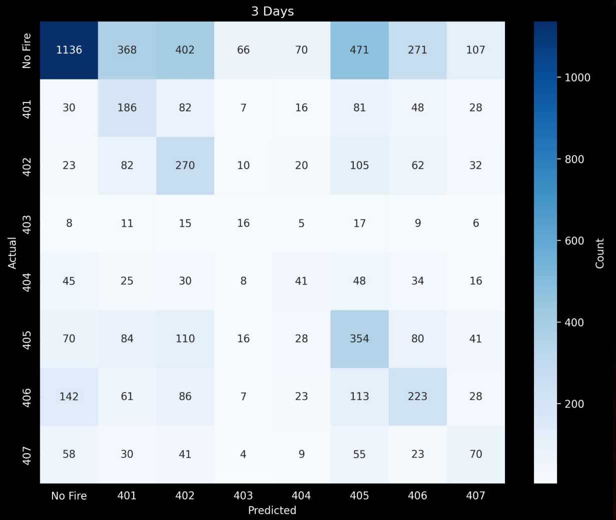
- Used auto ARIMA to iterate and determine optimal ordering values described below
- ARIMA order:
    - p: number of autoregressive (AR) terms
    - d: number of differencing terms
    - q: number of moving average (MA) terms
- Seasonal order:
    - P: seasonal AR
    - D: seasonal differencing
    - Q: seasonal MA
    - s: season step size

```
ARIMA order:  (1, 1, 2)
seasonal order:  (0, 0, 0, 7)
```

	Accuracy	Precision	Recall	F1
y_3	0.672	0.558	0.690	0.617
y_7	0.659	0.529	0.692	0.600
y_14	0.650	0.508	0.695	0.587

Insights

- ARIMA excels in seasonal forecasting with suspiciously little degradation in accuracy
- More time needed to investigate with other time-series models like Prophet
- Longer time periods needed to observe when dropoff in accuracy occurs



CONCLUSION

- Even when compared to a baseline random assignment probability of 12.5% across eight labels, the models delivers **strong short-term forecasting** (3-day), while still providing actionable intelligence for longer horizons despite increasing uncertainty.
- Model performance was impacted by **class imbalance** → majority of records indicated no fire occurrence
- Short-term predictions (3-day horizon) consistently achieved higher accuracy than longer-term forecasts (7- and 14-day), highlighting the increasing difficulty of capturing fire risk patterns over time
- **ARIMA** model yielded the best overall performance
- A key limitation of the dataset is the **scarcity of positive fire cases**, underscoring a broader challenge in wildfire research: the need for **more comprehensive and balanced fire incident data** to improve model training
- Future work will focus on exploring additional time series models, such as **LSTM networks** and **seasonal statistical approaches**, to better capture temporal dependencies and seasonal fire trends

REFERENCES

Original Dataset: Yavas, C. E., Kadlec, C., kim, J., & Chen, L. (2025). California Weather and Fire Prediction Dataset (1984–2025) with Engineered Features [Data set]. Zenodo. https://doi.org/10.5281/zenodo.14712845

Cal Fire Historical fire perimeters (1984 - 2023):
https://gis.data.ca.gov/datasets/CALFIRE-Forestry::california-fire-perimeters-all/explore?location=37.251628%2C-119.269051%2C5.96

Cal Fire Historical fire perimeters (2024-2025): https://lab.data.ca.gov/dataset/ca-perimeters-cal-fire-nifc-firis-public-view downloaded from
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CONUS Climate divisions boundaries https://psl.noaa.gov/data/usclimdivs/boundaries.html

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[https://corneliailin.github.io/datasci_w207_spring2025/summer2022_projects/Prakash_Krishnan_Final_Presentation_CA_Wildfires_Aug_4_2022.pdf]

THANKS!